

SATQUERY AI — REMOTE SENSING ANALYSIS REPORT

ISRO Problem Statement 26167 • Mission ID: e640c8be • Generated: 15:51:57.686Z

Natural Query:	Generate False-Color Infrared (CIR) composite to evaluate vegetation and water boundaries.
Classified Task:	BAND_COMPOSITE
Calibrated Confidence:	95% (Multi-Band Calibrated)
Specialist Models:	MultiSpectral-Composite-Synthesizer-v1
Total Latency:	129.1 ms

GROUNDING AI INTERPRETATION

A standard False-Color Infrared (Color-Infrared / CIR) composite was synthesized mapping Near-Infrared (NIR) to the Red channel, Red to Green, and Green to Blue. In this radiometric representation, healthy vegetative canopies (coastal scrub, casuarina plantations) appear brilliant ruby-red / crimson due to high cellular scattering in the 750-900nm infrared spectrum. Open water bodies (Pulicat Lake and marine perimeter) strongly absorb NIR and appear deep navy/black, while concrete launchpads, runways, and structural complexes reflect visible bands, appearing in steel-blue and cyan.

INPUT SENSOR METADATA

Filename	Modality	Dimensions	Bands	GeoTIFF CRS
isro_sdsc_optical.tif	OPTICAL	512x512	3	EPSG:4326

OBSERVABLE AGENTIC EXECUTION TRACE

#	Phase / Event	Observable Details	Time (ms)
1	Query Ingestion	Received natural language query: 'Generate False-Color Infrared (CIR) composite to evaluate vegetation and water boundaries.' with 1 uploaded image(s).	0.0
2	Input Validation	Verified 1 image file(s). Format and size limits satisfied.	0.0
3	Raster Metadata Extraction: isro_sdsc_optical.tif	Extracted dimensions (512x512), 3 band(s). CRS: EPSG:4326	76.1
4	Modality Identification	Detected sensor modalities: ['OPTICAL'].	0.0
5	Query Classification → BAND_COMPOSITE	Single image inquiry with multi-spectral band composite synthesis intent -> Routed to Multi-Spectral Composite Specialist (CIR/Agriculture).	0.0
6	Model Dispatch	Dispatched task to specialist: 'MultiSpectral-Composite-Synthesizer-v1' on device 'cpu'.	0.0

7	Specialist Execution Completed	Inference finalized in 52.3ms. Evidence artifacts compiled.	52.3
8	Result Fusion & Calibration	Derived grounded response with calibrated confidence 95% (Multi-Band Calibrated).	0.0